

Diqqat: .env faylni hech qachon git'ga commit qilmang. Loyihada .gitignore bu faylni allaqachon ignore qiladi.

3. config/settings.py da konfiguratsiya

Quyidagilar bu loyihada allaqachon yozilgan, lekin tushunish uchun ma'lumot sifatida keltirilmoqda.

3.1. INSTALLED_APPS

Quyidagi ikkita ilova ro'yxatda bo'lishi shart:

```
INSTALLED_APPS = [  
    # ...  
    'orders',  
    'payment',                      # ← bizning ilova  
    'telegrambot',  
    'paytechuz.integrations.django', # ← PayTechUZ  
]
```

3.2. PAYTECHUZ konfiguratsiyasi

Bu blok env-variables'dan o'qiydi — siz settings.py'ni o'zgartirmaysiz, faqat .env'dagi qiymatlarni o'zgartirasiz:

```
PAYTECHUZ = {  
    'PAYME': {  
        'PAYME_ID':      env('PAYME_ID', ''),  
        'PAYME_KEY':     env('PAYME_KEY', ''),  
        'ACCOUNT_MODEL': 'payment.models.Invoice',  
        'ACCOUNT_FIELD': 'id',  
        'AMOUNT_FIELD':  'amount',  
        'ONE_TIME_PAYMENT': True,  
        'IS_TEST_MODE':  env_bool('PAYME_TEST_MODE', True),  
    },  
    'CLICK': {  
        'SERVICE_ID':      env('CLICK_SERVICE_ID', ''),  
        'MERCHANT_ID':      env('CLICK_MERCHANT_ID', ''),  
        'MERCHANT_USER_ID': env('CLICK_MERCHANT_USER_ID', ''),  
        'SECRET_KEY':       env('CLICK_SECRET_KEY', ''),  
        'ACCOUNT_MODEL':    'payment.models.Invoice',  
        'ACCOUNT_FIELD':    'id',  
        'COMMISSION_PERCENT': 0.0,  
        'ONE_TIME_PAYMENT':  True,  
        'IS_TEST_MODE':      env_bool('CLICK_TEST_MODE', True),  
    },  
}
```

4. URL'larni ro'yxatlash

`config/urls.py` da quyidagi qator bo'lishi kerak:

```
urlpatterns = [
    path('admin/', admin.site.urls),
    path('accounts/', include('accounts.urls')),
    path('orders/', include('orders.urls')),
    path('payment/', include('payment.urls', namespace='payment')),
    path('reviews/', include('catalog.urls')),
    path('', include('store.urls')),
]
```

`payment/urls.py` ichida 3 ta endpoint registratsiya qilinadi:

```
app_name = 'payment'

urlpatterns = [
    path('create/', create_payment_link, name='create'),
    path('webhooks/payme/', PaymeWebhookView.as_view(), name='payme-webhook'),
    path('webhooks/click/', ClickWebhookView.as_view(), name='click-webhook'),
]
```

5. Endpoint'lar xulosasi

Endpoint	Maqsadi
POST /orders/submit/	Buyurtma yaratadi. Agar to'lov turi 'payme' yoki 'click' bo'lsa, javobda 'payment_url' qaytariladi.
POST /payment/create/	Mavjud buyurtma uchun to'lov havolasini yaratadi. Body: {order_code, provider, return_url}
POST /payment/webhooks/payme/	Payme tomonidan chaqiriladigan webhook. Merchant kabinetda ko'rsatiladi.
POST /payment/webhooks/click/	Click tomonidan chaqiriladigan webhook. Merchant kabinetda ko'rsatiladi.

6. Ma'lumotlar bazasini yangilash

Yangi **Invoice** modeli qo'shildi, hamda **Order.payment_method** tanlovi (Payme va Click alohida) o'zgartirildi. Migrationlarni qo'llang:

```
python manage.py makemigrations
python manage.py migrate
```

Yaratiladigan migrationlar:

- `orders/migrations/0003_alter_order_payment_method.py`
- `payment/migrations/0001_initial.py`

7. Merchant kabinetda webhook URL'larni ko'rsatish

Test rejimida ishlash uchun siz publik URL'ga ega bo'lishingiz kerak. Lokal mashinadan test qilish uchun **ngrok** yoki **cloudflared** ishlatishingiz mumkin:

```
ngrok http 8000
```

So'ngra olingan publik URL'ni quyidagi joylarda ko'rsating:

Payme business kabinet

Sozlama	Qiymat
Endpoint URL	https://<sizning-domain>/payment/webhooks/payme/
HTTP Method	POST
Account field	id (Invoice.id)

Click merchant kabinet

Sozlama	Qiymat
Prepare/Complete URL	https://<sizning-domain>/payment/webhooks/click/
HTTP Method	POST
Account field	id (Invoice.id)

8. Frontend'dan qanday foydalanish

8.1. Buyurtma yaratish bilan birga to'lov havolasi

submit_order javobida agar payment_method 'payme' yoki 'click' bo'lsa, **payment_url** avtomatik qaytariladi:

```
// frontend (savatdan checkout):
const res = await fetch('/orders/submit/', {
  method: 'POST',
  headers: {'Content-Type': 'application/json', 'X-CSRFToken': csrf},
  body: JSON.stringify({
    customer: {first_name, phone, address, ...},
    items: cartItems,
    delivery_method: 'standard',
    payment_method: 'payme', // yoki 'click'
  })
});
const data = await res.json();
if (data.payment_url) {
  window.location.href = data.payment_url;
}
```

8.2. Mavjud buyurtma uchun to'lov havolasi

Agar buyurtma allaqachon yaratilgan bo'lsa va keyinroq to'lov havolasi kerak bo'lsa:

POST /payment/create/
Content-Type: application/json

```
{  
  "order_code": "ORD-XXXXXXX",  
  "provider":   "payme",      // yoki "click"  
  "return_url": "https://sayt.uz/orders/"  // ixtiyoriy  
}
```

Javob:

```
{  
  "ok": true,  
  "invoice_id": 12,  
  "order_code": "ORD-XXXXXXX",  
  "provider":   "payme",  
  "amount":     "125000.00",  
  "payment_url": "https://checkout.paycom.uz/..."  
}
```

9. To'lov holatlari (status flow)

	Invoice.status	Order.status
	pending	new
li	paid (webhook)	confirmed (avtomatik)
r qildi yoki to'lov qaytarildi	cancelled (webhook)	(o'zgarmaydi)

Webhook chaqirilganda biz quyidagi mantiqni bajaramiz (**payment/views.py** ichida):

```
def _mark_paid(transaction):
    invoice = Invoice.objects.select_related('order').get(
        id=transaction.account_id
    )
    invoice.status = Invoice.STATUS_PAID
    invoice.save(update_fields=['status', 'updated_at'])
    order = invoice.order
    if order.status == Order.STATUS_NEW:
        order.status = Order.STATUS_CONFIRMED
        order.save(update_fields=['status', 'updated_at'])
```

10. Production'ga chiqarish checklist

- **.env** da **PAYME_TEST_MODE=false** va **CLICK_TEST_MODE=false**
- **.env** da merchant kabinetdan olingan haqiqiy (live) kalitlar
- **DJANGO_DEBUG=false**
- **SITE_URL** haqiqiy domain (https) bo'lishi shart — return URL'lar shu yerdan tortiladi
- **DJANGO_CSRF_TRUSTED_ORIGINS** ga haqiqiy domain qo'shing
- Merchant kabinetda webhook URL'lar publik HTTPS bo'lsin
- **python manage.py migrate** production bazasida ham bajarilgan
- Loglar (LOGGING) yoqilgan va xatolarni kuzatib boring (Sentry yoki shunga o'xshash)

Diqqat: webhook endpoint'lariga **CSRF** tegmaydi (PayTechUZ allaqachon csrf_exempt qiladi). Lekin to'lov tizimi tomonidan kelgan so'rovlarning haqiqiyliigi kalitlar orqali tekshiriladi — kalitlar maxfiy bo'lsin.

11. Tezkor lokal sinov

Server'ni ishga tushiring:

```
python manage.py runserver
```

Boshqa terminalda webhook'larni publik qiling:

```
ngrok http 8000
```

So'ng saytdan oddiy savat → checkout → Payme/Click tanlash bo'yicha buyurtma bering. Javobdagi **payment_url**'ga ergashing va test kartasi (Payme/Click hujjatlaridagi) bilan to'lov qiling. Webhook chaqirilgach Invoice **paid**, Order **confirmed** bo'lganini admin'da tekshiring:

```
http://127.0.0.1:8000/admin/payment/invoice/
```